

Electronics & Powertrain

WORKSHOP

2026

Duration	7 October 2026 – 4 November 2026 (4 weeks, 12 core sessions), followed by 2 complimentary sessions on 6 and 9 November 2026
Schedule	Monday, Wednesday and Friday
Timing	5:15 PM – 6:45 PM (1.5 hours per session) for all sessions
Venue	E6 Mechanical Seminar Hall (hands-on sessions: E6 Mechanical Seminar Hall & E6 CAD Lab; MATLAB session on 28 Oct: E3 207 EDA Centre & Power Systems Lab)
Audience	First-year students. No prior knowledge needed, we start from scratch!
Subjects	Network Analysis, Electronic Devices, Microcontrollers, Analog Circuits, Electric Machines & Motors, PCB Design
Format	8 talks, 4 hands-on sessions on the computer and 2 bonus sessions (Mechanical and Software)

Workshop Objectives

- Learn the basics of circuits, electronic parts, microcontrollers and motors, giving you a head start for your second semester and second year.
- See how the electronics inside the Team ASTERIX autonomous buggy actually work.
- Try things out yourself using circuit simulators, MATLAB and a PCB design tool.
- Get a peek at the mechanical and software side of the buggy in the bonus sessions.

1. Colour Legend

	Network Analysis
	Microcontrollers
	Electric Machines & Motors
	Hands-on / Simulation
	Industrial Expert Session

	Electronic Devices
	Analog Circuits
	PCB Design
	Complimentary Session

2. Timetable

Session Timing: 5:15 PM - 6:45 PM • **Industrial Expert Sessions (Saturdays): during college hours**

No.	Date	Subject	Topic	Handled by	Lecture Hall
Week 1 • 5 - 10 Oct					
1	7 Oct 2026 Wednesday	Network Analysis	Circuit Basics: How Electricity Flows	Kamaleshvar & Akshayaa	E6 Mechanical Seminar Hall
2	9 Oct 2026 Friday	Electronic Devices	Diodes & Transistors: The Tiny Switches	Dillimaran & Kathin	E6 Mechanical Seminar Hall
E1	10 Oct 2026 Saturday	Industrial Expert Session	<i>Industrial Expert Session During college hours</i>	Industry Expert (to be announced)	E8 Seminar Hall
Week 2 • 12 - 16 Oct					
3	12 Oct 2026 Monday	Hands-on Electronic Devices	Hands-on 1: Simulating a Transistor Switch	Handled jointly by the team	E6 Mechanical Seminar Hall & E6 CAD Lab
4	14 Oct 2026 Wednesday	Microcontrollers	Meet the Microcontroller: ESP32 Basics	Dr. Sujin (Dept of ECE)	E6 Mechanical Seminar Hall
5	16 Oct 2026 Friday	Analog Circuits	How Our Buggy Starts Up	Allwin & Akshayaa	E6 Mechanical Seminar Hall
Week 3 • 19 - 24 Oct					
-	19 Oct 2026 Monday	Pooja Holidays: Students can work on their assigned mini projects			
6	21 Oct 2026 Wednesday	Hands-on Analog Circuits / Microcontrollers	Hands-on 2: Build the Buggy's Start-Up in Tinkercad	Handled jointly by the team	E6 Mechanical Seminar Hall & E6 CAD Lab
7	23 Oct 2026 Friday	Analog Circuits	Power Up & Clean Up: Buck-Boost Converters, Op-Amps & Filters	Dr. Sathiyathan (Dept of Robotics and Autonomous Systems)	E6 Mechanical Seminar Hall
E2	24 Oct 2026 Saturday	Industrial Expert Session	<i>Industrial Expert Session During college hours</i>	Industry Expert (to be announced)	E8 Seminar Hall
Week 4 • 26 - 31 Oct					
8	26 Oct 2026 Monday	Electric Machines & Motors	How Electric Motors Work	Aravind & Vishal	E6 Mechanical Seminar Hall
9	28 Oct 2026 Wednesday	Hands-on Electric Machines & Motors	Hands-on 3: Simulating Electric Motors in MATLAB	Dr. Ravikrishna (Dept of EEE)	E3 207 EDA Centre & Power Systems Lab
10	30 Oct 2026 Friday	PCB Design	PCB Design I: What Is a PCB and How Is It Made?	Saswin & Nanthavikraman	E6 Mechanical Seminar Hall
E3	31 Oct 2026 Saturday	Industrial Expert Session	<i>Industrial Expert Session During college hours</i>	Industry Expert (to be announced)	E8 Seminar Hall
Week 5 • 2 - 7 Nov (final core sessions + complimentary session)					
11	2 Nov 2026 Monday	PCB Design	PCB Design II: Laying Out a Board	Saswin & Nanthavikraman	E6 Mechanical Seminar Hall
12	4 Nov 2026 Wednesday	Hands-on PCB Design	Hands-on 4: Design Your First PCB	Saswin & Nanthavikraman	E6 Mechanical Seminar Hall & E6 CAD Lab
C1	6 Nov 2026 Friday	Complimentary Session	Complimentary Session: Mechanical Basics	Team ASTERIX	E6 Mechanical Seminar Hall
E4	7 Nov 2026 Saturday	Industrial Expert Session	<i>Industrial Expert Session During college hours</i>	Industry Expert (to be announced)	E8 Seminar Hall
Week 6 • 9 Nov (complimentary session + valedictory)					
C2	9 Nov 2026 Monday	Complimentary Session	Complimentary Session: Software Basics + Valedictory	Team ASTERIX	E6 Mechanical Seminar Hall

3. Session-by-Session Guide

SESSION 1 · NETWORK ANALYSIS

Circuit Basics: How Electricity Flows

Date	Wednesday, 7 Oct 2026	Time	5:15 PM - 6:45 PM
Handled by	Kamaleshvar & Akshayaa	Type	Lecture
Lecture Hall	E6 Mechanical Seminar Hall		

Topics covered

- The basics: voltage, current, resistance and power, and how they relate
- Ohm's law and combining resistors in series and parallel
- Kirchhoff's laws (KCL & KVL) to solve simple circuits
- How capacitors charge and discharge in RC circuits, and where we use them

By the end, you'll be able to:

- ✓ Use Ohm's law and Kirchhoff's laws on simple circuits
- ✓ Explain how a capacitor charges and discharges

SESSION 2 · ELECTRONIC DEVICES

Diodes & Transistors: The Tiny Switches

Date	Friday, 9 Oct 2026	Time	5:15 PM - 6:45 PM
Handled by	Dillimaran & Kathin	Type	Lecture
Lecture Hall	E6 Mechanical Seminar Hall		

Topics covered

- What semiconductors are and how a diode lets current flow only one way
- How transistors (BJT and MOSFET) work
- Using a MOSFET as an electronic switch
- Why MOSFETs are the go-to choice in power circuits

By the end, you'll be able to:

- ✓ Explain what a diode does
- ✓ Use a MOSFET as a simple on/off switch

SESSION 3 · HANDS-ON · ELECTRONIC DEVICES

Hands-on 1: Simulating a Transistor Switch

Date	Monday, 12 Oct 2026	Time	5:15 PM - 6:45 PM
Handled by	Handled jointly by the team	Type	Hands-on (Simulation)
Lecture Hall	E6 Mechanical Seminar Hall & E6 CAD Lab		

Topics covered

- Getting started with PSpice / LTspice: placing parts and wiring a circuit
- Building a simple MOSFET test circuit
- Running simulations and plotting how the MOSFET behaves
- Bonus: simulating a MOSFET switching a load on and off

By the end, you'll be able to:

- ✓ Build and run a basic circuit simulation
- ✓ Read a graph that shows how a MOSFET behaves

SESSION 4 · MICROCONTROLLERS

Meet the Microcontroller: ESP32 Basics

Date	Wednesday, 14 Oct 2026	Time	5:15 PM - 6:45 PM
Handled by	Dr. Sujin	Type	Lecture + Demo
Lecture Hall	E6 Mechanical Seminar Hall		

Topics covered

- What a microcontroller is and why modern vehicles are full of them
- Meet the ESP32: its main features and pins
- Setting up Arduino IDE and writing your first program
- First projects: blinking an LED, reading a button and dimming an LED

By the end, you'll be able to:

- ✓ Name the main parts of a microcontroller like the ESP32
- ✓ Upload your first program and make an LED blink

Before the session: Install Arduino IDE. Create a free Tinkercad account for the hands-on session on 21 Oct.

SESSION 5 · ANALOG CIRCUITS

How Our Buggy Starts Up

Date	Friday, 16 Oct 2026	Time	5:15 PM - 6:45 PM
Handled by	Allwin & Akshayaa	Type	Lecture
Lecture Hall	E6 Mechanical Seminar Hall		

Topics covered

- A tour of the buggy's electrical system, from battery to loads
- The starting sequence: how the buggy powers up, step by step
- Safety and kill switches that keep everything under control
- How small control signals switch big loads, plus lessons from competition

By the end, you'll be able to:

- ✓ Walk through the buggy's start-up steps
- ✓ Explain how a small signal can switch on a big load

SESSION 6 · HANDS-ON · ANALOG CIRCUITS / MICROCONTROLLERS

Hands-on 2: Build the Buggy's Start-Up in Tinkercad

Date	Wednesday, 21 Oct 2026	Time	5:15 PM - 6:45 PM
Handled by	Handled jointly by the team	Type	Hands-on (Simulation)
Lecture Hall	E6 Mechanical Seminar Hall & E6 CAD Lab		

Topics covered

- Getting to know Tinkercad Circuits and the Arduino Uno
- Building the buggy's starting sequence with buttons, LEDs and a motor
- Coding the sequence, including safety checks and an emergency stop
- Testing and debugging everything in simulation

By the end, you'll be able to:

- ✓ Code a simple start-up sequence with safety checks
- ✓ Test your circuit and code in simulation

SESSION 7 · ANALOG CIRCUITS

Power Up & Clean Up: Buck-Boost Converters, Op-Amps & Filters

Date	Friday, 23 Oct 2026	Time	5:15 PM - 6:45 PM
Handled by	Dr. Sathiyathan	Type	Lecture
Lecture Hall	E6 Mechanical Seminar Hall		

Topics covered

- Part 1 · Power Up: why a vehicle needs different voltage levels for its electronics
- Linear vs switching regulators: which one wastes less energy
- How buck (step-down) and boost (step-up) converters work, and the ready-made modules used in the buggy
- Part 2 · Clean Up: what an op-amp is and the simple rules it follows
- Common op-amp circuits: amplifiers, buffers and comparators
- Why sensor signals need filtering, and how simple low-pass and high-pass filters work

By the end, you'll be able to:

- ✓ Explain how buck and boost converters step voltage down or up
- ✓ Recognise basic op-amp circuits and explain how a simple filter removes unwanted noise

SESSION 8 · ELECTRIC MACHINES & MOTORS

How Electric Motors Work

Date	Monday, 26 Oct 2026	Time	5:15 PM - 6:45 PM
Handled by	Aravind & Vishal	Type	Lecture
Lecture Hall	E6 Mechanical Seminar Hall		

Topics covered

- How electricity and magnetism work together to make motors spin
- DC motors and how we control their speed and direction
- BLDC, stepper and servo motors, and where each one is used
- The motor and controller used in our buggy

By the end, you'll be able to:

- ✓ Explain how a motor turns electricity into motion
- ✓ Tell apart DC, BLDC, stepper and servo motors

SESSION 9 · HANDS-ON · ELECTRIC MACHINES & MOTORS

Hands-on 3: Simulating Electric Motors in MATLAB

Date	Wednesday, 28 Oct 2026	Time	5:15 PM - 6:45 PM
Handled by	Dr. Ravikrishna	Type	Hands-on (Simulation)
Lecture Hall	E3 207 EDA Centre & Power Systems Lab		

Topics covered

- Getting started with MATLAB and Simulink: the workspace, blocks and building a simple model
- Building a DC motor model from its basic parts: voltage, resistance, inductance and back EMF
- Watching the motor start up: how speed and current change over time when voltage is applied
- Changing the supply voltage and the load to see how speed and current respond
- Plotting the motor's characteristics: speed vs torque and current vs torque
- Comparing the simulation with the motor used in our buggy

By the end, you'll be able to:

- ✓ Build and run a simple motor model in MATLAB / Simulink
- ✓ Read and explain a motor's speed, current and torque graphs

SESSION 10 · PCB DESIGN

PCB Design I: What Is a PCB and How Is It Made?

Date	Friday, 30 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Saswin & Nanthavikraman	Type	Lecture + Demo
Lecture Hall	E6 Mechanical Seminar Hall		

Topics covered

- What a PCB is and why we use it instead of a breadboard
- Parts of a PCB: tracks, pads, the green coating and the white labels
- Through-hole vs surface-mount components
- The design journey from circuit diagram to finished board, with a software demo

By the end, you'll be able to:

- ✓ Identify the main parts of a PCB
- ✓ Describe, in simple steps, how a circuit diagram becomes a finished board

SESSION 11 · PCB DESIGN

PCB Design II: Laying Out a Board

Date	Monday, 2 Nov 2026	Time	5:15 PM – 6:45 PM
Handled by	Saswin & Nanthavikraman	Type	Lecture + Demo
Lecture Hall	E6 Mechanical Seminar Hall		

Topics covered

- Turning a circuit diagram into a board layout
- Placing parts neatly and drawing tracks the right way
- Checking the design for mistakes before the board is made
- A tour of the PCB inside the Team ASTERIX buggy

By the end, you'll be able to:

- ✓ Follow the basic rules for placing parts and drawing tracks
- ✓ Point out the main sections of the buggy's PCB

SESSION 12 · HANDS-ON · PCB DESIGN

Hands-on 4: Design Your First PCB

Date	Wednesday, 4 Nov 2026	Time	5:15 PM – 6:45 PM
Handled by	Saswin & Nanthavikraman	Type	Hands-on (Guided Demonstration)
Lecture Hall	E6 Mechanical Seminar Hall & E6 CAD Lab		

Topics covered

- Follow along as the instructors design a simple LED board
- Drawing the circuit and placing the parts
- Connecting the tracks, running the design check and fixing mistakes
- Seeing the finished board in 3D, plus Q&A

By the end, you'll be able to:

- ✓ Follow the complete design process for a very simple circuit

SESSION C1 · COMPLIMENTARY SESSION**Complimentary Session: Mechanical Basics**

Date	Friday, 6 Nov 2026	Time	5:15 PM – 6:45 PM
Handled by	Team ASTERIX	Type	Complimentary Session
Lecture Hall	E6 Mechanical Seminar Hall		

Topics covered

- A look at the buggy's mechanical parts: frame, suspension, steering, brakes and gears
- How the electronics and the mechanical parts work together

By the end, you'll be able to:

- ✓ See how the mechanical and electrical parts of the buggy fit together

SESSION C2 · COMPLIMENTARY SESSION**Complimentary Session: Software Basics + Valedictory**

Date	Monday, 9 Nov 2026	Time	5:15 PM – 6:45 PM
Handled by	Team ASTERIX	Type	Complimentary Session + Valedictory
Lecture Hall	E6 Mechanical Seminar Hall		

Topics covered

- A look at the software that drives the autonomous buggy, plus programming basics
- Valedictory: workshop wrap-up, feedback and closing

By the end, you'll be able to:

- ✓ Understand what software does in the autonomous buggy

4. Subject-wise Summary

Subject	Session s	Dates	Hours
Network Analysis	1	7 Oct	1.5 h
Electronic Devices	1	9 Oct	1.5 h
Microcontrollers	1	14 Oct	1.5 h
Analog Circuits	2	16, 23 Oct	3.0 h
Electric Machines & Motors	1	26 Oct	1.5 h
PCB Design	2	30 Oct, 2 Nov	3.0 h
Hands-on / Simulation	4	12 Oct (PSPICE/LTSPICE), 21 Oct (Tinkercad), 28 Oct (MATLAB), 4 Nov (PCB)	6.0 h
Complimentary Session	2	6 Nov (Mechanical), 9 Nov (Software + Valedictory)	3.0 h
Total	14	12 core sessions + 2 complimentary sessions	21.0 h

Student Timetable

Electronics & Powertrain Workshop 2026 · Team ASTERIX · aBAJA SAEINDIA
All sessions: 5:15 PM – 6:45 PM · **Industrial Expert Sessions (Saturdays): during college hours**

Week	Monday	Wednesday	Friday	Saturday
Week 1 5 – 10 Oct	Mon 5 Oct <i>No session — workshop begins Wednesday</i>	Wed 7 Oct NETWORK ANALYSIS Circuit Basics: How Electricity Flows <i>Kamaleshvar & Akshayaa</i> E6 Mechanical Seminar Hall	Fri 9 Oct ELECTRONIC DEVICES Diodes & Transistors: The Tiny Switches <i>Dillimaran & Kathin</i> E6 Mechanical Seminar Hall	Sat 10 Oct INDUSTRIAL EXPERT SESSION During college hours <i>Industry Expert (to be announced)</i> E8 Seminar Hall
Week 2 12 – 16 Oct	Mon 12 Oct HANDS-ON · ELECTRONIC DEVICES Simulating a Transistor Switch <i>Handled jointly by the team</i> E6 Mechanical Seminar Hall & E6 CAD Lab	Wed 14 Oct MICROCONTROLLERS Meet the Microcontroller: ESP32 Basics <i>Dr. Sujin (Dept of ECE)</i> E6 Mechanical Seminar Hall	Fri 16 Oct ANALOG CIRCUITS How Our Buggy Starts Up <i>Allwin & Akshayaa</i> E6 Mechanical Seminar Hall	Sat 17 Oct <i>No session</i>
Week 3 19 – 24 Oct	Mon 19 Oct POOJA HOLIDAYS Students can work on their assigned mini projects	Wed 21 Oct HANDS-ON · ANALOG CIRCUITS / MICROCONTROLLERS Build the Buggy's Start-Up in Tinkercad <i>Handled jointly by the team</i> E6 Mechanical Seminar Hall & E6 CAD Lab	Fri 23 Oct ANALOG CIRCUITS Power Up & Clean Up: Buck-Boost Converters, Op-Amps & Filters <i>Dr. Sathiyathan (Dept of Robotics and Autonomous Systems)</i> E6 Mechanical Seminar Hall	Sat 24 Oct INDUSTRIAL EXPERT SESSION During college hours <i>Industry Expert (to be announced)</i> E8 Seminar Hall
Week 4 26 – 31 Oct	Mon 26 Oct ELECTRIC MACHINES & MOTORS How Electric Motors Work <i>Aravind & Vishal</i> E6 Mechanical Seminar Hall	Wed 28 Oct HANDS-ON · ELECTRIC MACHINES & MOTORS Simulating Electric Motors in MATLAB <i>Dr. Ravikrishna (Dept of EEE)</i> E3 207 EDA Centre & Power Systems Lab	Fri 30 Oct PCB DESIGN PCB Design I: What Is a PCB and How Is It Made? <i>Saswin & Nanthavikraman</i> E6 Mechanical Seminar Hall	Sat 31 Oct INDUSTRIAL EXPERT SESSION During college hours <i>Industry Expert (to be announced)</i> E8 Seminar Hall
Week 5 2 – 7 Nov	Mon 2 Nov PCB DESIGN PCB Design II: Laying Out a Board <i>Saswin & Nanthavikraman</i> E6 Mechanical Seminar Hall	Wed 4 Nov HANDS-ON · PCB DESIGN Design Your First PCB <i>Saswin & Nanthavikraman</i> E6 Mechanical Seminar Hall & E6 CAD Lab	Fri 6 Nov COMPLIMENTARY SESSION Mechanical Basics <i>Team ASTERIX</i> E6 Mechanical Seminar Hall	Sat 7 Nov INDUSTRIAL EXPERT SESSION During college hours <i>Industry Expert (to be announced)</i> E8 Seminar Hall
Week 6 9 Nov	Mon 9 Nov COMPLIMENTARY SESSION Software Basics + Valedictory <i>Team ASTERIX</i> E6 Mechanical Seminar Hall			

Network Analysis	Electronic Devices	Microcontrollers	Analog Circuits	Electric Machines & Motors	PCB Design	Hands-on / Simulation	Complimentary Session	Industrial Expert Session
------------------	--------------------	------------------	-----------------	----------------------------	------------	-----------------------	-----------------------	---------------------------

Lecture Hall: E6 Mechanical Seminar Hall; hands-on sessions (12, 21 Oct and 4 Nov) in E6 Mechanical Seminar Hall & E6 CAD Lab; MATLAB session (28 Oct) in E3 207 EDA Centre & Power Systems Lab; Industrial Expert Sessions (Saturdays) in E8 Seminar Hall. Bring a laptop with the required software installed to the hands-on sessions.

5. Mini Project Sample Ideas (Pooja Holidays)

Simulation-only projects, no hardware needed, for students to work on during the Pooja holidays (19 Oct). They use only what is taught up to 16 Oct, and every project is themed around a real vehicle system.

5.1 In LTspice

No.	Project	What to build	Concepts used
1	Battery Health Monitor	A vehicle's 12 V battery must be read by a 3.3 V controller. Design a voltage divider that scales 9–14 V into a safe range, sweep the battery voltage and mark the “full”, “low” and “dead” levels on the graph.	<i>Voltage divider, Ohm's law</i>
2	Turn Indicator Flasher	Use a pulse source to drive a MOSFET that switches an indicator lamp (modelled as a resistor). Set it to blink about 1.5 times per second, close to a real car.	<i>MOSFET as a switch, timing</i>
3	Reverse-Polarity Protection	Simulate the battery connected correctly and backwards, with and without a protection diode. Calculate the voltage and power the diode loses.	<i>Diode forward/reverse bias, power</i>
4	Horn / Fuel-Pump Relay Driver	Drive a relay coil (resistor + inductor) with a MOSFET. Capture the voltage spike when it switches off, then add a flyback diode to remove it.	<i>MOSFET switch, flyback diode</i>
5	Headlight Dimmer	Use PWM on a MOSFET to run a headlight LED at 100% (high beam) and 30% (low beam). Compare the average current in each case.	<i>PWM, power saving</i>

5.2 In Tinkercad (Arduino Uno)

No.	Project	What to build	Concepts used
1	Indicator & Hazard Lights	Left and right switches blink the matching LEDs; a hazard button blinks both. Make sure left and right can never blink together.	<i>Digital inputs and outputs, logic</i>
2	Seatbelt & Door Warning	The car “starts” (green LED) only if the door is closed and the seatbelt is on (both switches). Otherwise a buzzer beeps and a warning LED lights.	<i>Safety interlocks</i>
3	Reverse Parking Assist	An ultrasonic sensor at the “rear” makes a buzzer beep faster as an obstacle gets closer, with green / yellow / red LEDs for the distance zone. Stop the motor automatically when too close.	<i>Reading sensors, decision-making in code</i>
4	Throttle & Cruise Control	A potentiometer acts as the accelerator and sets motor speed through PWM and a transistor. A brake button cuts the motor instantly; a cruise button holds the current speed until the brake is pressed.	<i>Analog input, PWM, transistor switch, safety logic</i>

Stretch ideas for faster students

- Combine the Seatbelt & Door Warning with Throttle & Cruise Control into a mini “drive-by-wire” car: it cannot move unless the safety checks pass, and the brake always overrides the throttle.
- Add automatic headlights to any Tinkercad project using an LDR, so the lights turn on when it gets dark.

How to submit

- Work in pairs and pick at least one project.
- Submit the LTspice .asc file or a Tinkercad share link, a screenshot of the result, and a 5-line explanation of how it works.
- Tinkercad simulates the Arduino Uno, not the ESP32. The code is almost identical, so it can be reused on an ESP32 later.